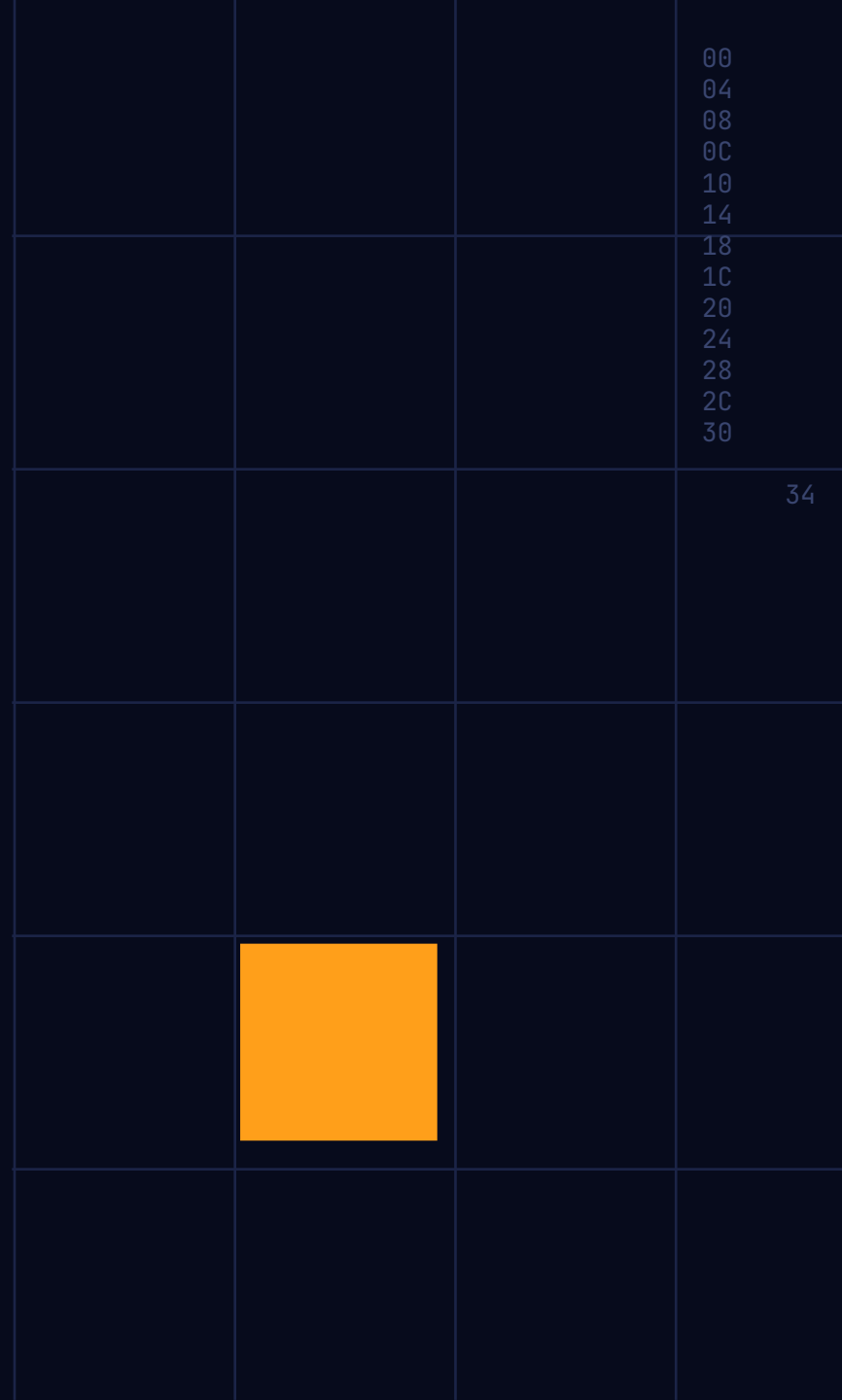
A SHORT HISTORY OF MUSIC GENERATION ON COMPUTERS

MAKING MUSIC ON COMPUTERS.

From OctaMED on the Amiga to ChipWave Studio — how the tracker taught home computers to sing.

1987 → 2025 · MOD · S3M · XM · IT · CWT



Computers sang before trackers

1951

CSIRAC, Melbourne

The first computer known to play music hums “Colonel Bogey” through a loudspeaker.

1982

The SID chip

The Commodore 64 puts a real synthesizer — three voices, filters, envelopes — into a home computer.

1983

MIDI

One cable, one language: keyboards, drum machines and computers finally talk to each other.

1985

Amiga + Paula

Four channels of sampled 8-bit sound — the hardware the tracker was invented for.

Sources: Wikipedia — CSIRAC · MOS Technology 6581 · MIDI · Amiga

1987

THE ULTIMATE SOUNDTRACKER · KARSTEN OBARSKI

Commercially it flopped. As an idea, it went everywhere. Obarski's program arranged short samples into scrolling **patterns**: four channels, one for each voice of the Amiga's Paula sound chip.

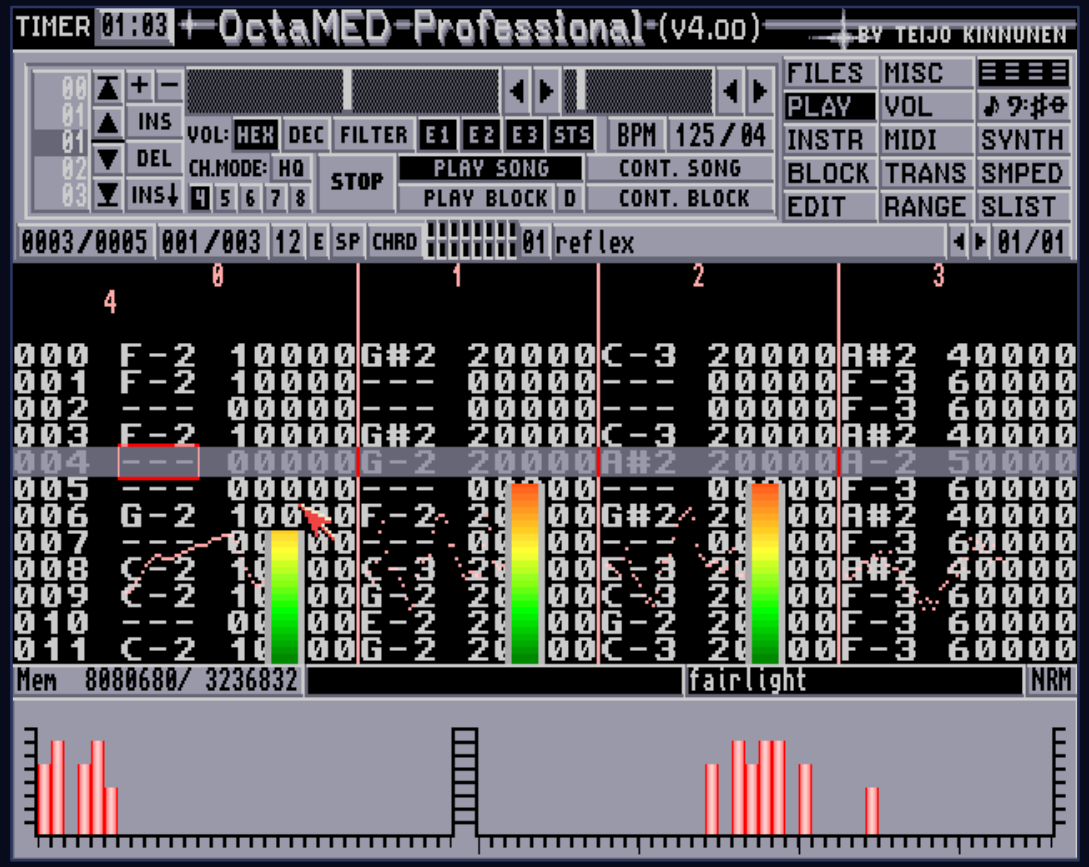
Cracked, shared and cloned within months, it became the **MOD format** — free music software before free music software had a name, and the seed of the demoscene. Every tracker since, OctaMED included, is built on this grid.

ROW	CH 1	CH 2	CH 3	CH 4
00	C-3 01 000	--- -- ---	G-2 02 000	--- -- ---
01	--- -- ---	--- -- ---	--- -- ---	--- -- ---
02	E-3 01 000	--- -- ---	--- -- ---	--- -- ---
03	--- -- ---	--- -- ---	A#2 02 000	--- -- ---
04	G-2 01 C40	C-3 03 000	--- -- ---	F-2 04 F03
05	--- -- ---	--- -- ---	--- -- ---	--- -- ---
06	C-3 01 000	--- -- ---	G-2 02 000	--- -- ---
07	--- -- ---	D-3 03 000	--- -- ---	--- -- ---
08	A#2 01 000	--- -- ---	--- -- ---	A#2 04 000
09	--- -- ---	--- -- ---	--- -- ---	--- -- ---

A tracker pattern: note · instrument · effect, per channel.

Source: Wikipedia — Ultimate Soundtracker · MOD (file format)

OctaMED squeezes eight channels out of four



OctaMED Professional v4 on the Amiga · shot: nostalgicplayer.dk

SOFTWARE MIXING

Teijo Kinnunen's MED (Music EDitor, 1989) became OctaMED in 1991: two software-mixed voices in each of Paula's four hardware channels — eight tracks on a stock Amiga.

MIDI FROM THE START

Unlike its rivals, MED drove external synths over the serial port — one Amiga could sequence a rack of studio gear and play its own samples at the same time.

THE SOUND OF JUNGLE

Aphrodite, Bizzy B, Omni Trio and DJ Zinc cut early drum & bass on it. Calvin Harris made *I Created Disco* on an A1200 — and brought the Amiga on stage at the BRIT Awards 2024.

Source: Wikipedia — OctaMED

The tracker explosion

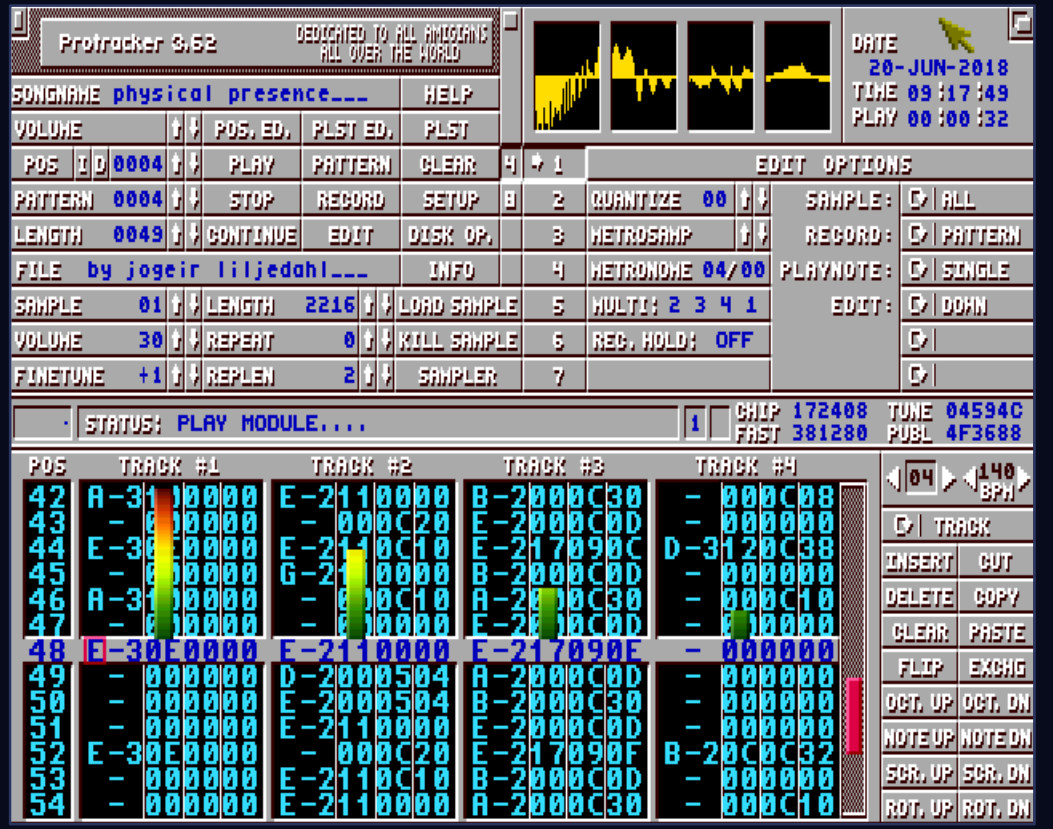
1990 — **ProTracker** — free, fast and fixed; the Amiga scene standard that took MOD worldwide.

1994 — **Scream Tracker 3** — the PC joins in; S3M powers a wave of demoscene classics.

1994 — **FastTracker II** — XM adds instruments, envelopes and up to 32 channels.

1995 — **Impulse Tracker** — IT pushes the PC to 64 channels with resonant filters.

MOD → S3M → XM → IT



ProTracker 3.52 on the Amiga · image: Wikipedia

Sources: Wikipedia — ProTracker · Scream Tracker · FastTracker 2 · Impulse Tracker

The timeline replaces the pattern list.

As PCs grew faster, arranging moved to the piano roll and the horizontal timeline: record, comp, automate. Trackers never died — they went back underground, kept alive by the demoscene and a stubborn few who thought in rows, not bars.

1996 · CUBASE VST

Software instruments and effects move into the box.

1997 · FRUITYLOOPS

Step sequencing and loops for everyone, on any PC.

2001 · ABLETON LIVE

Clips and scenes turn arranging into performance.

2002 · RENOISE

The tracker, modernised — the grid survives the DAW era.

Chiptune: limitation becomes style

The constraint that forced the tracker's invention turns into an aesthetic of its own.

2000	**LSDJ** — a cartridge turns the Nintendo Game Boy into a four-channel groovebox.
2005	**FamiTracker** — composing for the NES sound chip, free on the PC.
2012	**DefleMask** — one tracker for a dozen classic chips, Mega Drive to C64.
2021	**Furnace** — open source, and every sound chip you remember, in one grid.

Square waves at parties. Pattern rows on stage. The tracker was never nostalgia — it stayed a working tool.

ChipWave Studio — a free studio with a retro heart



ChipWave Studio 0.2.18, the Timeline and piano roll · real capture, chipwavestudio.com

FREE & OPEN SOURCE

GPL-3.0. No account, no telemetry, no subscription. Windows, macOS and Linux, written in Rust.

ONE WINDOW

Timeline and clip launcher, piano roll, drum kit, 32 tracks, a mixer with a true-peak limiter — and, yes, a pattern editor.

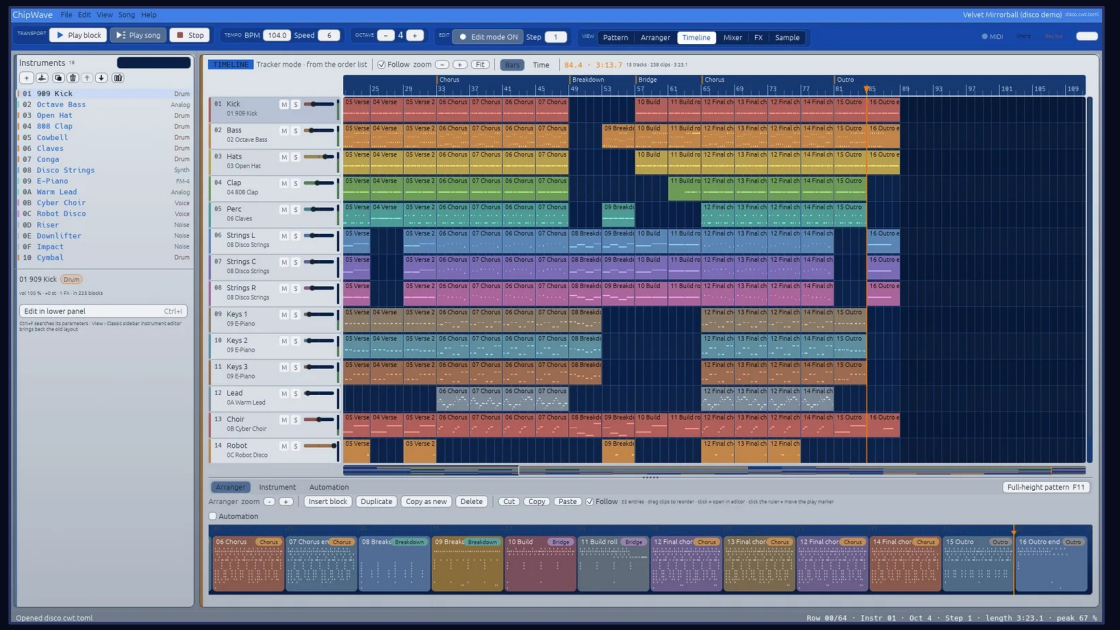
SOUNDS FROM SCRATCH

WaveForge, StringWave, FM-4, Analog and Chip FX, plus CLAP and VST3 plugins. Renders WAV and MP3.

The old way is still in there



The Pattern Editor: OctaMED-style blocks, note and effect columns.



The Amiga theme: Workbench blue title bars, orange cursor.

BLOCKS & COLUMNS — 64-row blocks, the Z–M keyboard, selection, transpose and named undo — the 1991 workflow, intact.

TRACKER EFFECTS — Slides, decays, tempo and filter commands — plus 303 glide and accent (3C / 3D) for acid lines.

CHIP FX — An sfxr-style generator: pickups, lasers and blips that play in tune with the pattern note.

1987 → TODAY · SAME GRID, SAME IDEA

The pattern still scrolls.

Four hardware channels in 1987. Thirty-two tracks, synths and a true-peak limiter today. The pattern editor didn't get replaced — it grew up.

DOWNLOAD · chipwavestudio.com

SOURCE · github.com/postmanpat-303/ChipWaveStudio

FREE · GPL-3.0 · NO ACCOUNT · WINDOWS / MACOS / LINUX

